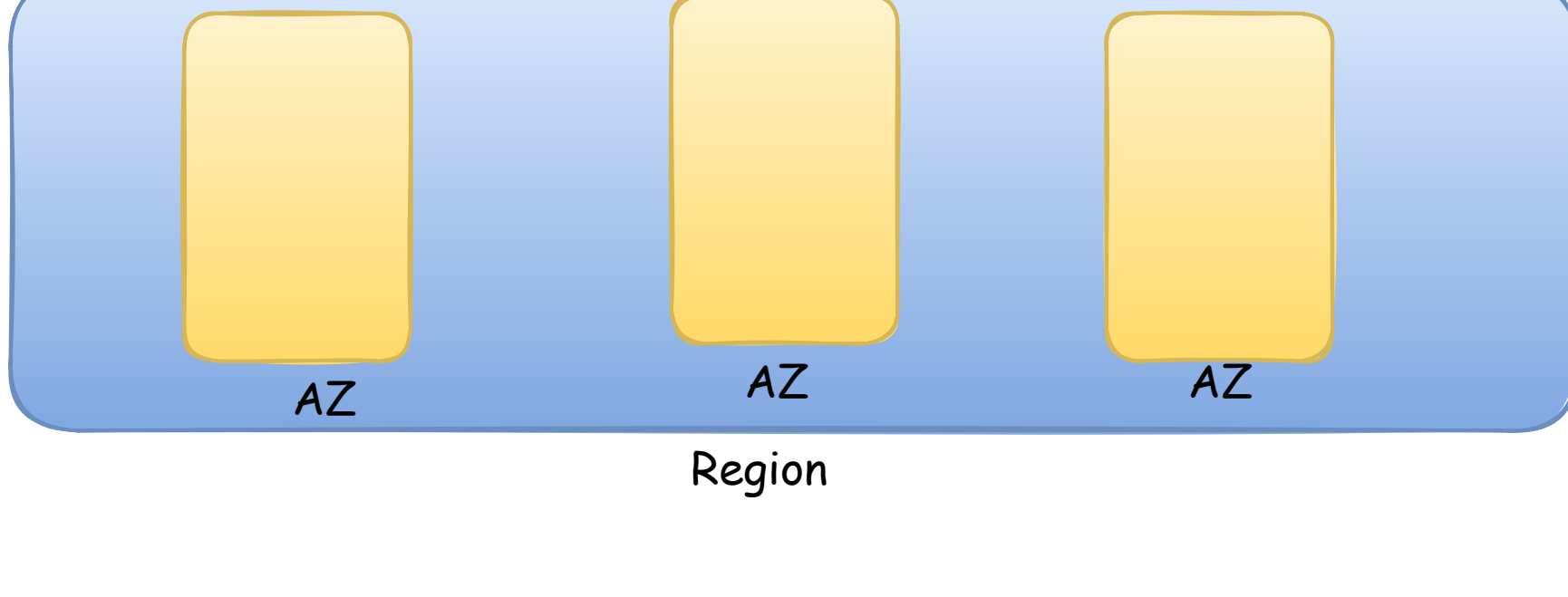
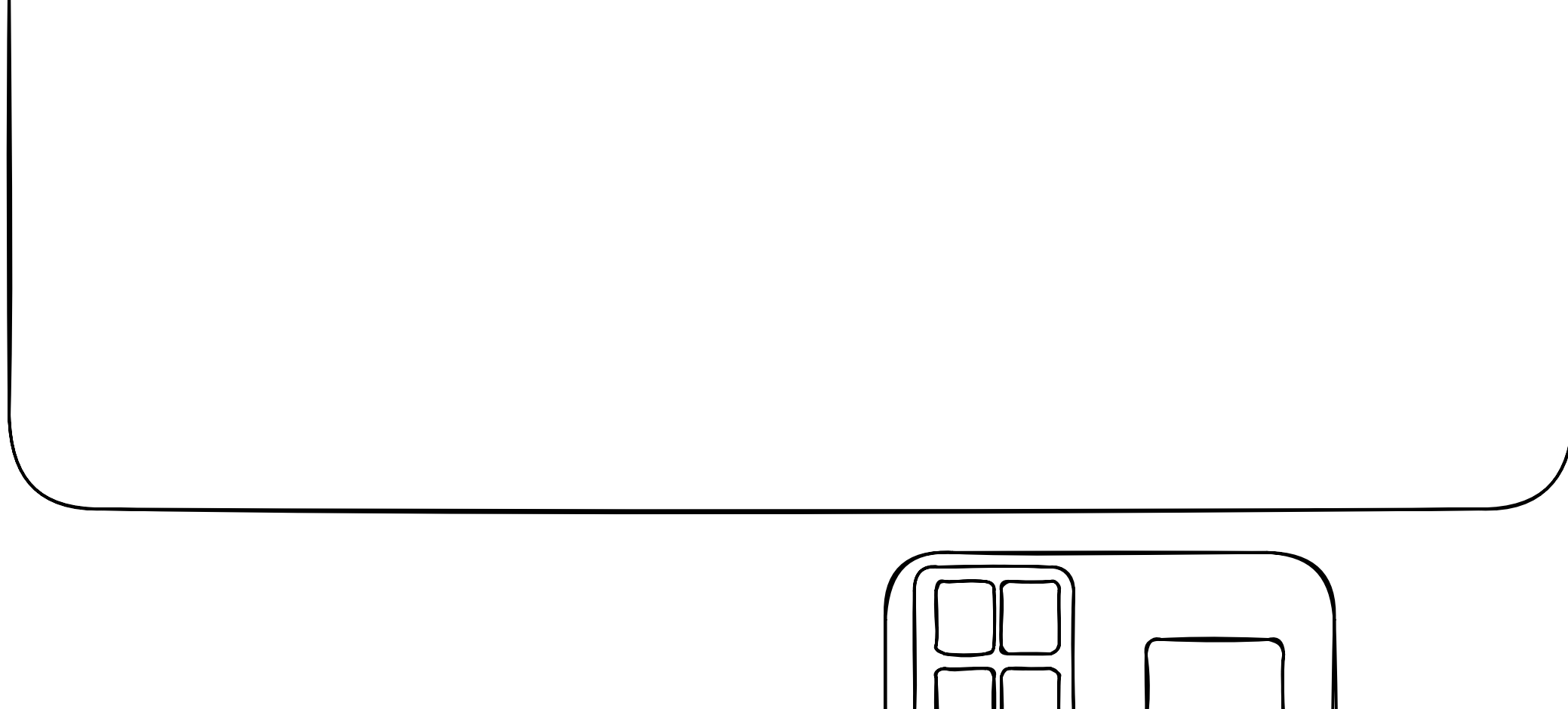


## AWS VPC



ap-south-1 → ap-south-1a → ap-south-1b → ap-south-1c  
 ↓  
 ap-south-2

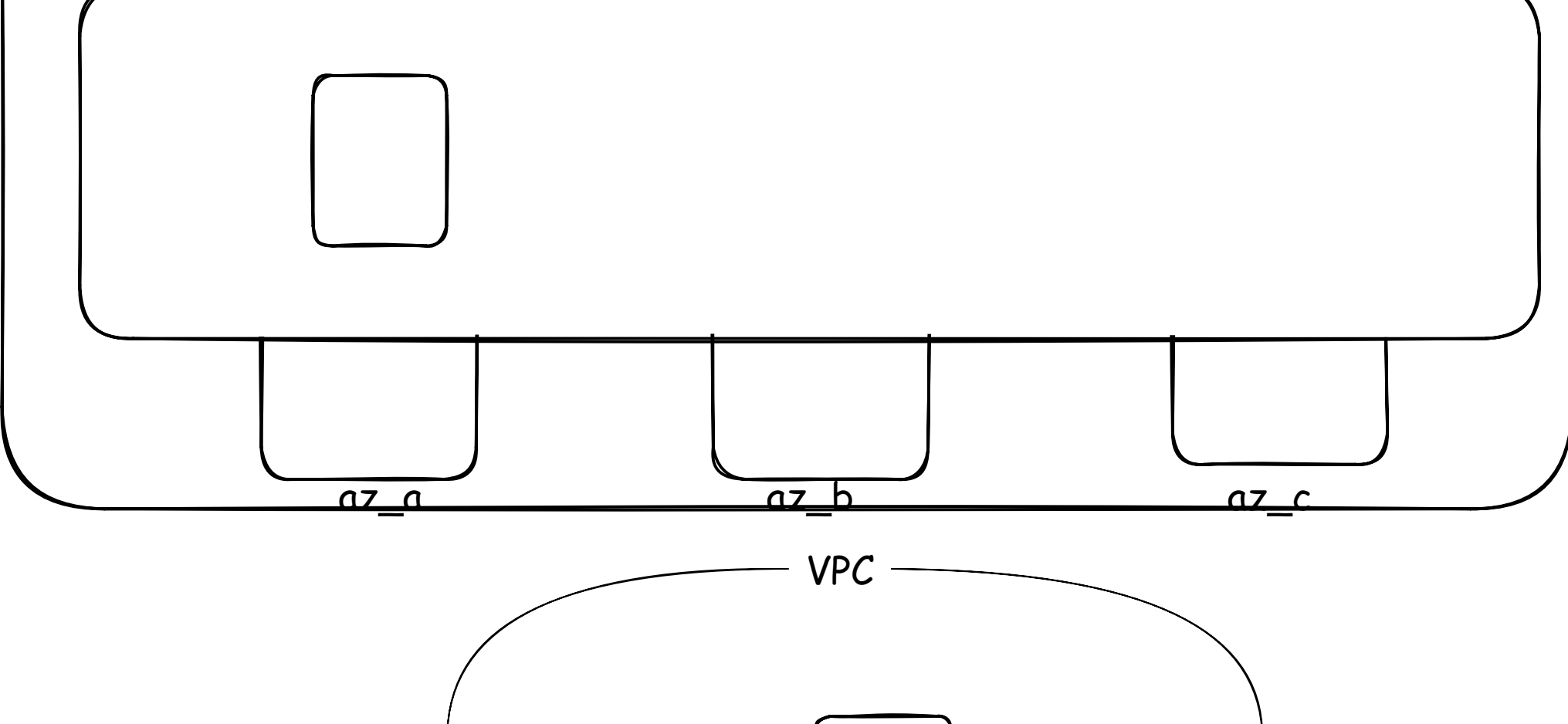
AWS --- email → root Account



AWS is PUBLIC Cloud  
 Private Cloud

Virtual Private Cloud  
 Default VPC

Own VPC → Custom VPC

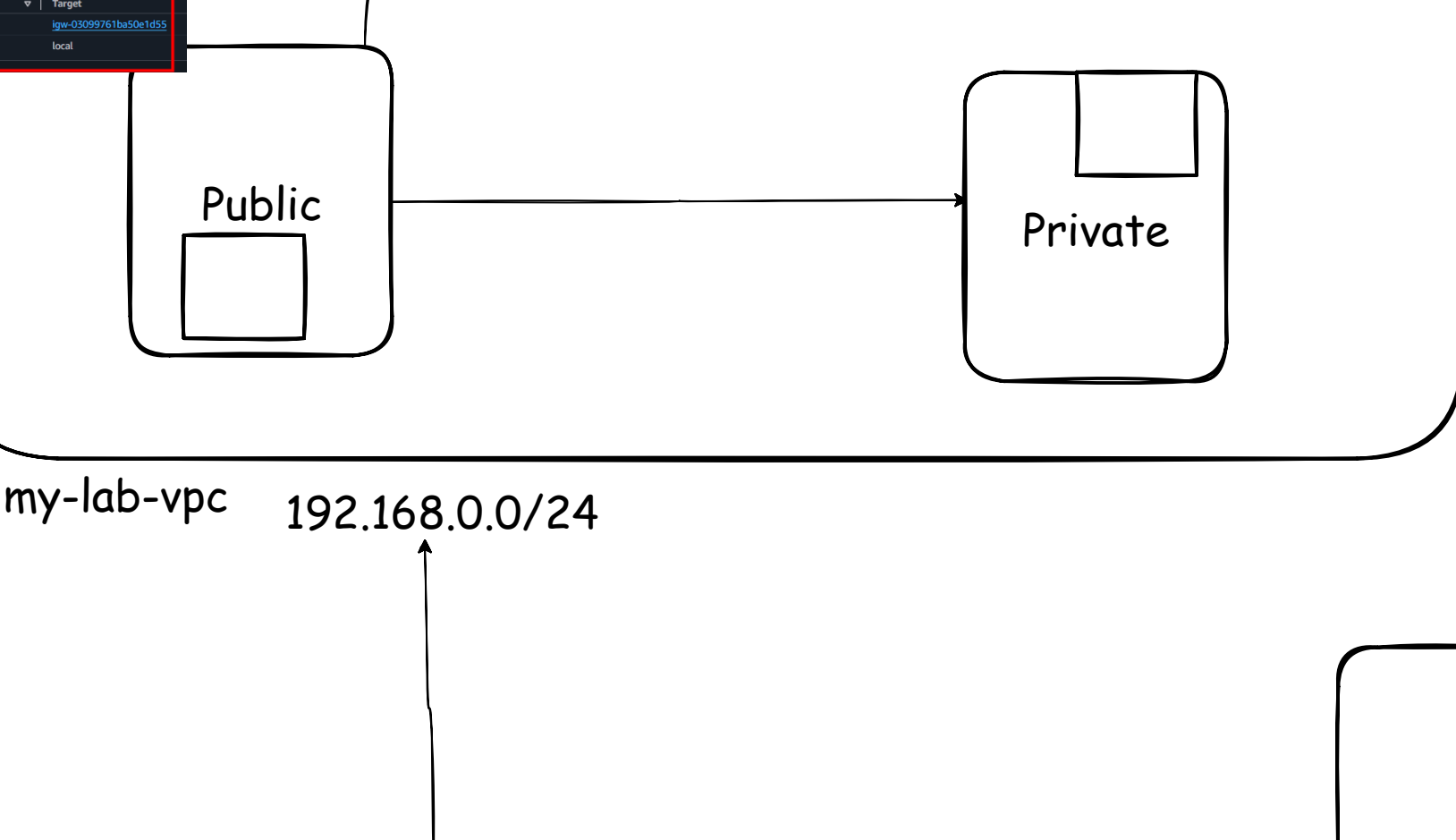


VPC

Name

CIDR

IGW



my-lab-vpc

192.168.0.0/24

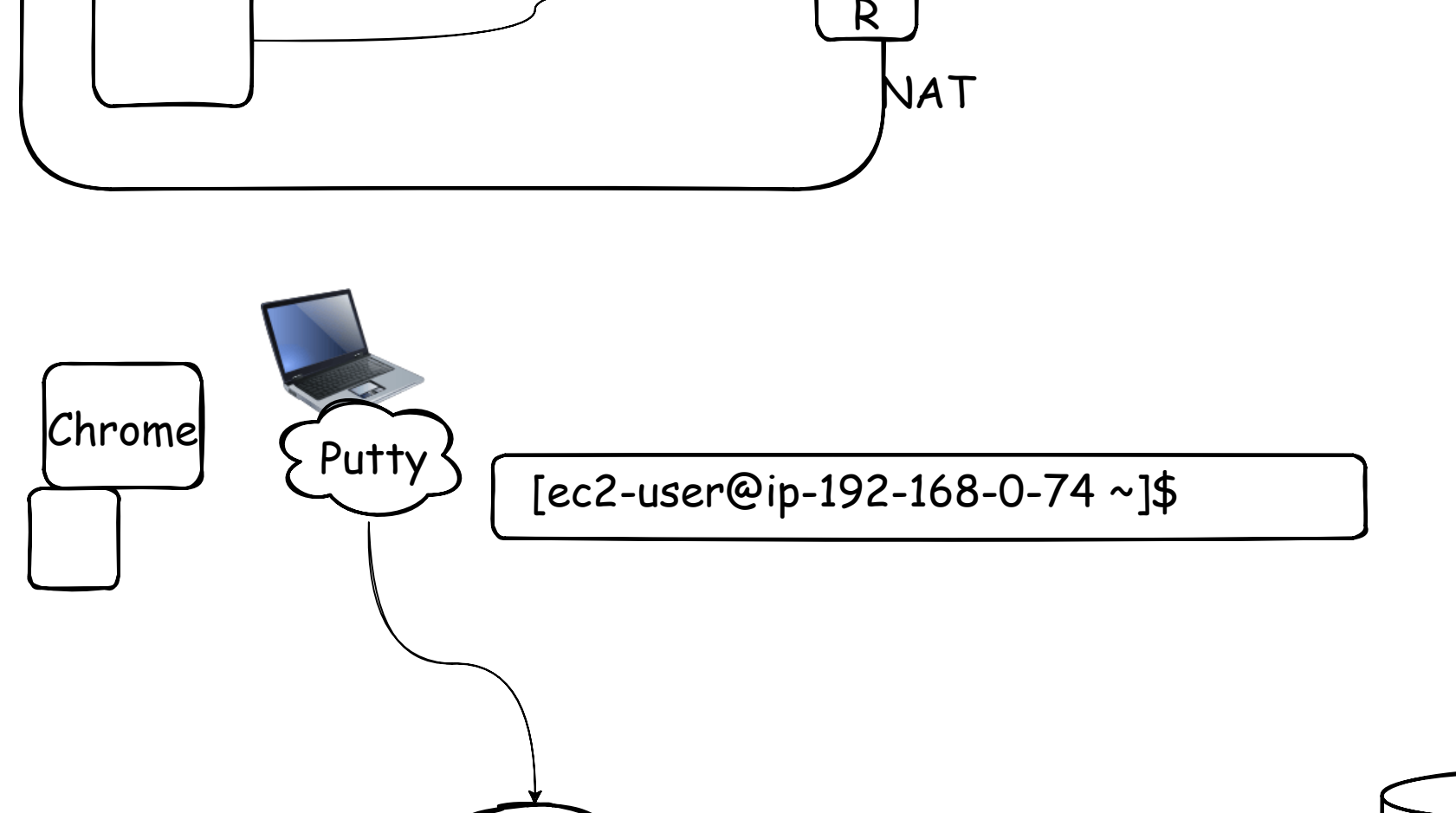
N/W Admin

Public Means -----> I want to access INTERNET

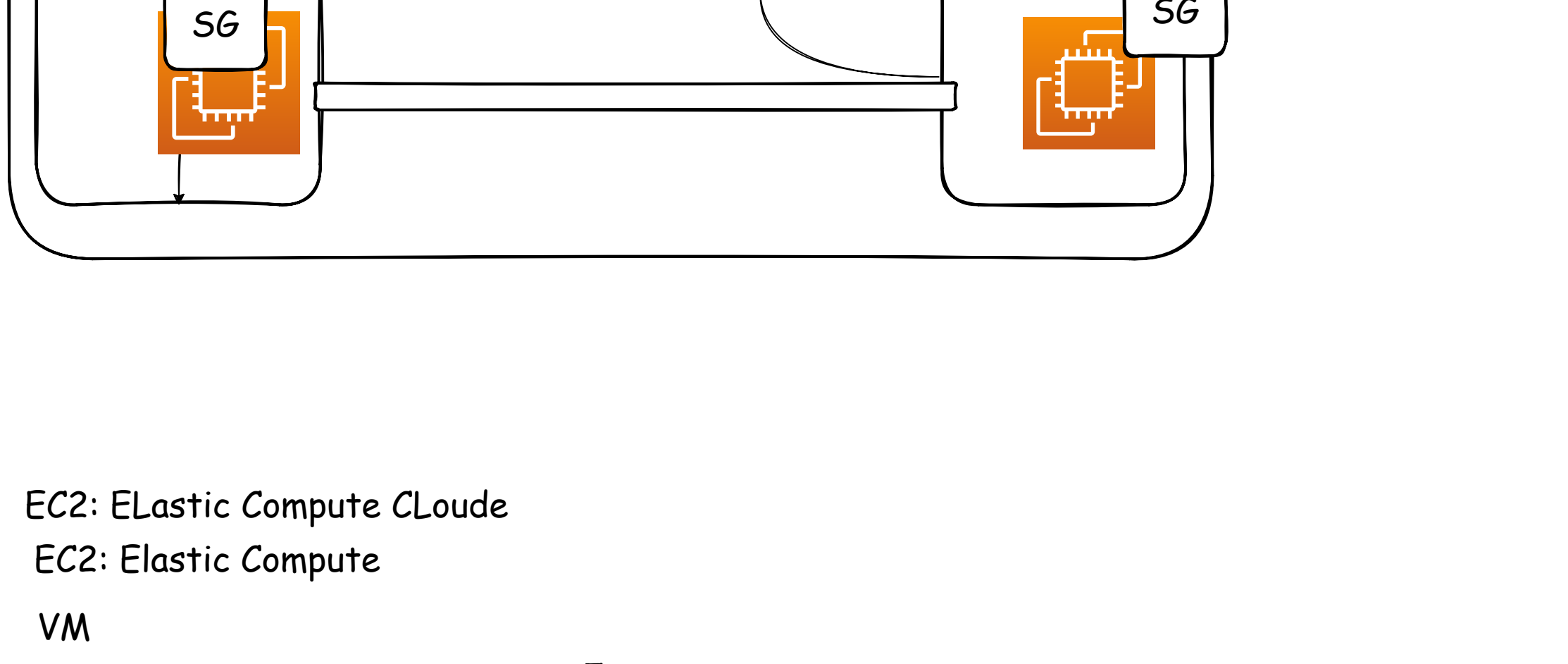
Any amchine wants to tals to INTERNET

#1: Must have public IP Address

#2: Musthave ROUTES towards INTERNET



NGW



EC2: ELastic Compute CLoude

EC2: ELastic Compute

VM



A RT can associated with many Subnets

A Sub

AWS FireWall

Security group

NACLs

Attach to EC2 INstance

NetworkAccess Control List

Ststeless FireWall

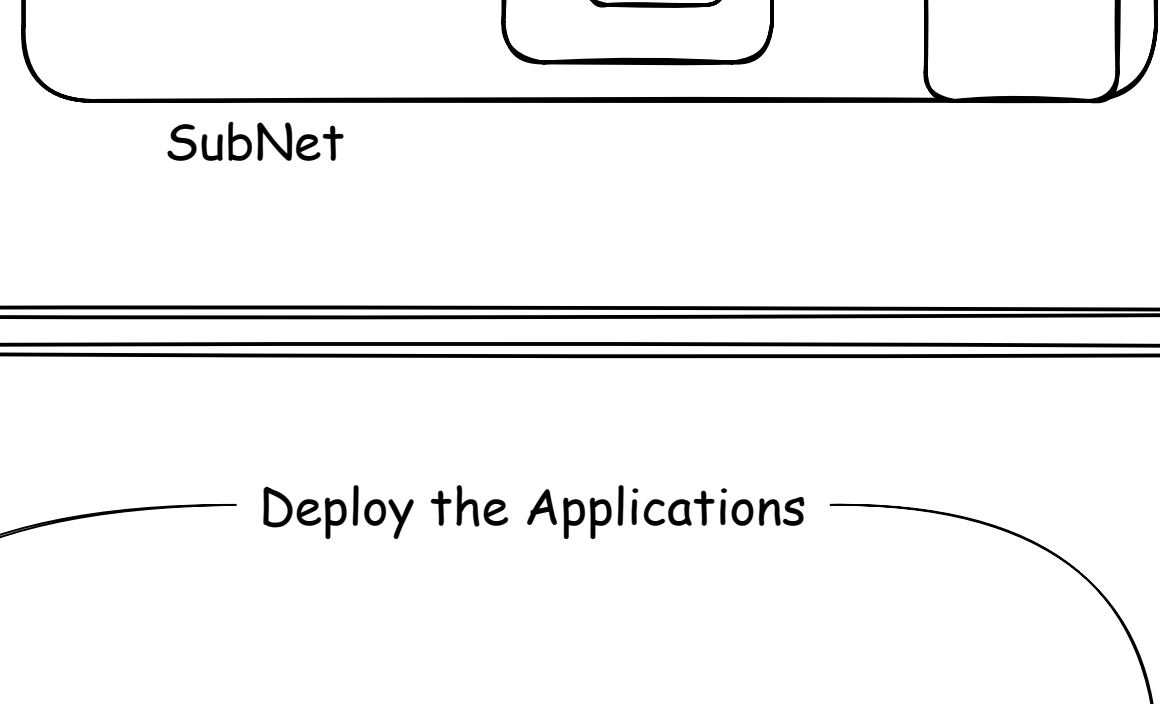
works at Subnet Level

We don't bother about outBound Traffic

StateLess InBound outBound

NACLs

NACLd: The rules have number lower r the number higher the prisdence



Day-2

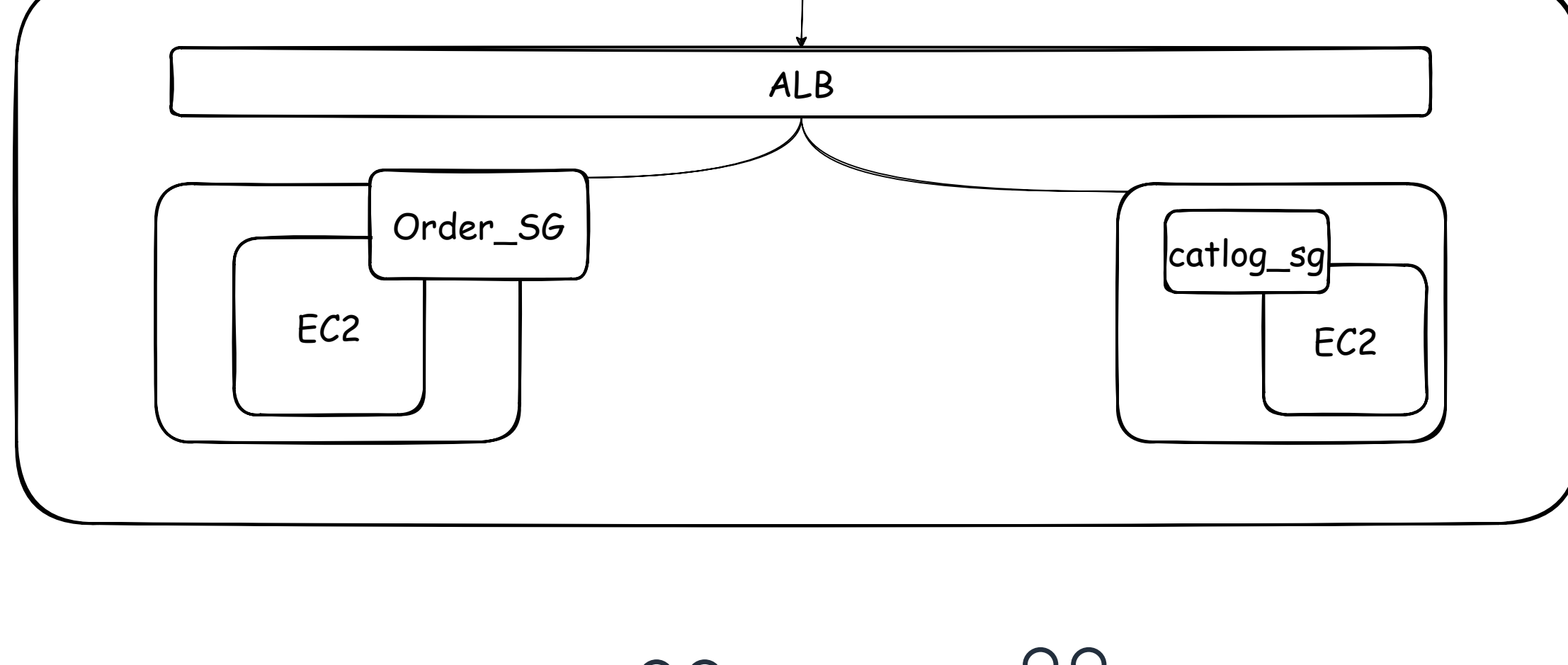
Deploy the Applications

Deploy catalog-api

Deploy orders-api

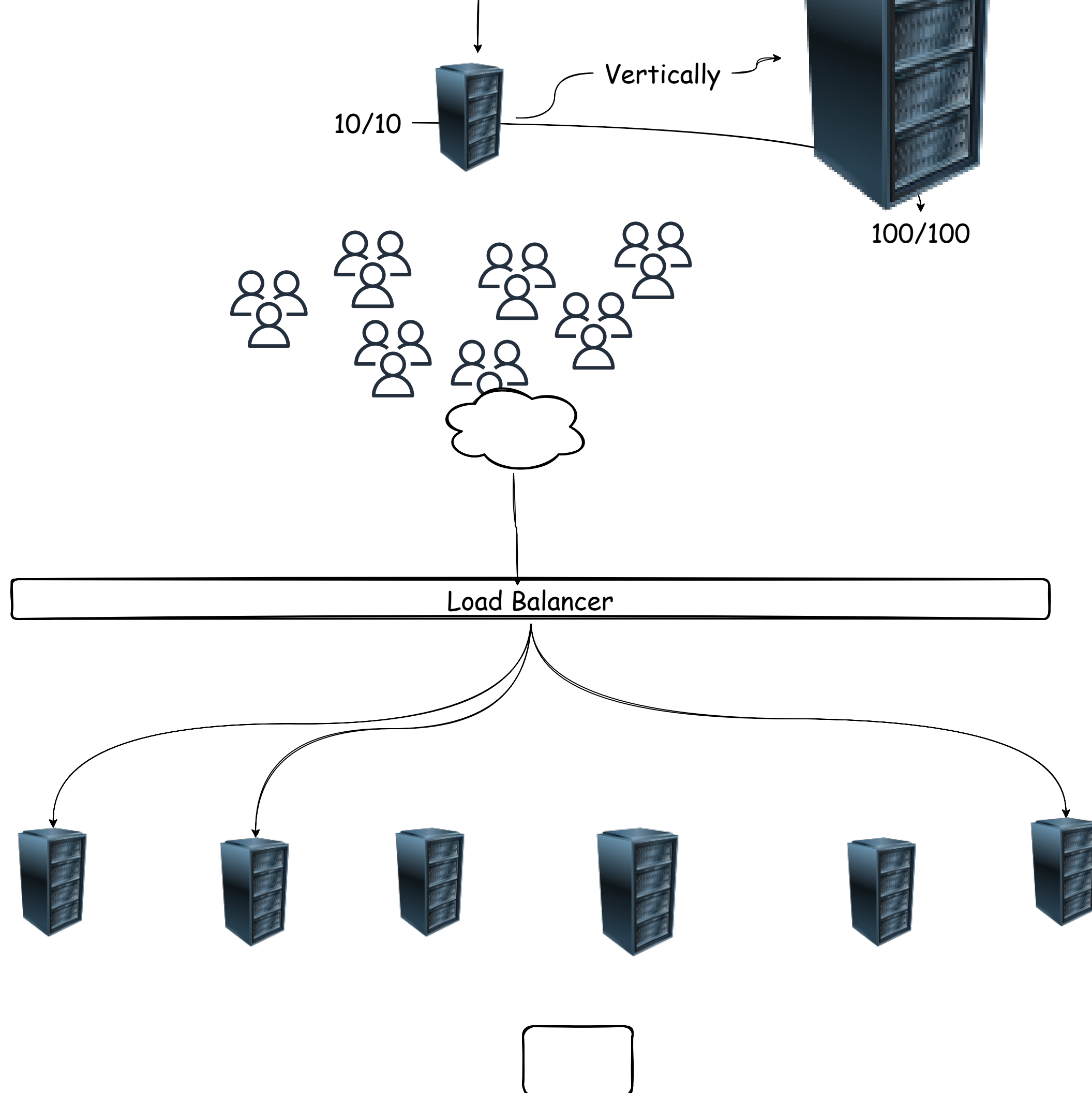
Configure systemd

Configure systemd



Scaling

DayTime



ASG

Golden

Launch Confiuration

AMI

AMI

Golden

Install App + Config

LoadBalancer

HealthCheck

Listener

BackEnd

Green

LoadBalancer will send the traffic

Port on which LoadBalancer Listen

The Application Servers

OSI Model

APPLICATION

PRESENTATION

SESSION

TRANSPORT

NETWORK

DATA LINK

PHYSICAL

Types of LB

ALB Application Load Balancer

NLB: Network Load Balancer

Layer 7 LB

Classic LB

Layer 4 LB

It understand HTTP/HTTPS

It works at 7 and 4 both

TCP/UDP

| Use ALB when                         | Use NLB when                                                             |
|--------------------------------------|--------------------------------------------------------------------------|
| Routing by URL path or hostname      | Traffic is not HTTP — MQTT, databases, game protocols                    |
| Terminating HTTPS certificates       | You need a fixed IP address to give a partner for firewall allow-listing |
| Different services behind one domain | You need extreme throughput with minimal added latency                   |
| You want request-level logging       | You must preserve the client IP without HTTP headers                     |